L9.2 Modern SLAM for AD Quiz

1. A pinhole camera has f = 800 px and principal point cx = 640 px. A point at X = 2 m, Z = 10 m (camera frame) projects to which horizontal pixel u?
A) u = 640
B) u = 160
C) u = 800
D) u = 1,440

2. Why can a single (monocular) camera not recover metric scale?
A) Scaling the whole scene and the camera translation by the same factor produces identical images
B) Monocular cameras have no lens distortion model
C) The focal length is always unknown in a monocular camera
D) Feature matching fails when only one camera is used

3. What do ORB features consist of, and how are descriptors compared?
A) Gradient histograms of 128 floats, compared with the Euclidean distance
B) Raw pixel patches, compared with the cross-correlation of the whole image
C) Learned 3-D point descriptors, compared with the Mahalanobis distance
D) Oriented FAST corners with rotated BRIEF binary descriptors, compared with the Hamming distance

4. What does the ratio test do in feature matching?
A) It removes all features whose descriptor has more than half of its bits set
B) It rejects a match if the best descriptor distance is not clearly smaller than the second best (ambiguous match)
C) It computes the ratio between the image width and height to rescale the features
D) It keeps only the match with the largest descriptor distance

5. RANSAC with sample size s = 3 and inlier ratio w = 0.5 needs about how many iterations for 99 % confidence? (N = log(1 − p) / log(1 − w^s))
A) About 35
B) About 3
C) About 8
D) About 1,200

6. What does the epipolar constraint $x_2^T E x_1 = 0$ with E = [t]× R allow us to estimate?
A) The absolute position of the camera in the world frame, with metric scale
B) The depth of every pixel of the first image
C) The relative rotation R and the direction of the translation t (but not its length) between two views
D) The intrinsic parameters of the camera (f, cx, cy)

7. In ORB-SLAM tracking, what problem does PnP (Perspective-n-Point) solve?
A) The 3-D position of a point from two known camera poses
B) The camera pose from known 3-D map points and their 2-D pixel observations
C) The loop-closure candidate with the most similar bag of words
D) The scale of a monocular map from the IMU

8. What does bundle adjustment minimize?
A) The Hamming distance between all pairs of ORB descriptors
B) The time between consecutive camera frames
C) The number of keyframes stored in the map
D) The (robust) sum of squared reprojection errors over camera poses and 3-D map points

9. Which set describes the main parts of ORB-SLAM3?
A) Tracking, local mapping, and loop closing / map merging threads, plus an Atlas of multiple maps
B) A detector, a tracker, and a planner
C) Odometry at 10 Hz, mapping at 1 Hz, and no loop closure
D) A camera front-end, an EKF back-end with N landmarks, and particles

10. What does adding an IMU to visual SLAM mainly provide?
A) Dense color for every map point
B) Removal of the need for camera calibration
C) Metric scale and gravity direction, and the ability to bridge short visual failures such as blur
D) Immunity to GNSS spoofing
11. In LOAM, the smoothness value c of a LiDAR point is computed from its neighbors on the scan line. How is it used?
A) A large c marks a planar feature; a small c marks an edge feature
B) A large c marks an edge feature (matched to lines); a small c marks a planar feature (matched to planes)
C) c is the reflectivity of the point and is used for place recognition
D) c is the time stamp of the point and is used to deskew the scan
12. What is the idea of the dual-rate design of LOAM?
A) Two LiDARs running at different frequencies, fused with a Kalman filter
B) Two separate maps, one for edges and one for planes, merged at the end
C) A camera at 30 Hz and a LiDAR at 10 Hz, synchronized by the IMU
D) A fast, coarse scan-to-scan odometry (about 10 Hz) plus a slower, more accurate scan-to-map mapping step
13. Which description matches LIO-SAM?
A) A pure EKF with N landmarks and range-bearing measurements
B) A bag-of-words visual place recognition method
C) A factor graph with IMU preintegration, LiDAR odometry, optional GNSS, and loop-closure factors, solved incrementally with iSAM2
D) A particle filter that samples the LiDAR scan
14. Why does scan matching fail in a long, straight corridor?
A) All plane normals are perpendicular to the corridor axis, so $H = \Sigma\, n\, n^T$ has a zero eigenvalue along the corridor
B) The LiDAR cannot measure distances to walls
C) The walls reflect too much light and saturate the sensor
D) ICP always needs at least 1,000 points to converge
15. A lane is 3.7 m wide and the car is 1.9 m wide. If 3σ of the lateral localization error must stay within the one-sided margin, what is the largest σ?
A) About 0.9 m
B) About 0.3 m
C) About 1.8 m
D) About 0.05 m
16. Which recipe do Autoware and Apollo both follow for localization in a prior map?
A) Pure visual SLAM with a monocular camera and no prior map
B) GNSS only, because RTK is always available
C) A neural network that outputs the pose directly, with no filter
D) IMU/odometry propagation, LiDAR scan matching against the map, and GNSS correction in a Kalman filter
17. An attacker adds a GNSS bias that grows by 0.12 m per step. Why does a per-step innovation (χ^2) gate barely notice it?
A) Each step's disagreement is within the noise, so the filter absorbs the bias and follows the spoofer; a windowed check is needed
B) The χ^2 test is only valid for LiDAR measurements
C) The gate threshold is always larger than 100 m
D) The attacker removes the innovation from the filter's memory
18. What is the main lesson of the FusionRipper attack on multi-sensor fusion (Shen et al., 2020)?
A) Multi-sensor fusion makes GPS spoofing impossible
B) The attack only works if the car has no IMU

C) Fusing several sensors is necessary but not sufficient: the fusion logic itself can be exploited by GPS spoofing to make the pose drift
D) The attack requires physical access to the LiDAR

19. What does SLAMSpoof target, and what did it achieve in real-world tests?

A) The camera exposure; it blinded the car for one second
B) LiDAR-based scan matching in localization; position errors of 4.2 m or more against three popular algorithms
C) The HD map file on disk; it deleted lane markings
D) The wheel encoders; it changed the odometry scale by 1 %

20. A localization pose arrives 80 ms late while the car drives at 25 m/s. How far does the car travel in that time?

A) 0.2 m
B) 0.32 m
C) 2.0 m
D) 20 m